VIRTUAL INTERNSHIP AT CISCO-AICTE

Submitted in partial fulfilment of requirement for degree of Bachelor of technology

IN

COMPUTER SCIENCE ENGINEERING

MEERUT INSTITUTE OF ENGINEERING & TECHNOLOGY, MEERUT



Submitted by

ARSHAD KHAN

1900680100074

Index

S.NO	TITLE	PAGENO.
1	Index	2
2	Acknowledgement	3
3	Description	4
4	Introduction to Cyber Security	5
5	Cybersecurity Essential	8
6	Introduction to packet tracer	12
7	Overview	14
8	My project	16

Acknowledgement

I am grateful to the CISCO Networking Academy For allowing me to study **CYBER SECURITY.**

In the time of pandemic, they provide us virtual internship. While the pandemic has shaken up the traditional internship model, the opportunities for work experience provided by CISCO.

In collaboration with AICTE this program was initiative to build skill sets for the future.

The aim of this internship is to be familiar to the practical aspect and uses of theoretical knowledge in the field of cyber security.

I am also like express my sincere gratitude to Meerut Institute of Engineering & Technology, our internship/placement coordinator and all faculty who have given their valuable time and given me chance to learn something new.

Name: - Arshad Khan

Roll No.: - 1900680100074

Description

Cisco Networking Academy delivers a comprehensive learning experience to help students develop ICT skills for career opportunities, continuing education, and globally recognized career certifications.

Cisco in collaboration with AICTE and NASSCOM launched Virtual Internship Program for us which aimed at building key skill sets for the future. This unique collaboration between industry and government helped us to build a skilled talent pool for tech jobs in the industry.

In this program I was enrolled with 3 different courses,

-
- **Introduction to Cyber Security**: - Learned what cybersecurity is and its potential impact to us.
- **Cyber Essentials:** - Learn valuable security principals and how to develop compliant policies.
- **Introduction to Packet Tracer: -** Practice networking, IoT, and cybersecurity skills virtually with Cisco Packet Tracer, a powerful simulation and visualization tool.

“Cybersecurity talent demand continues to outpace supply and is expected to increase five times over the next four years in India. To bridge this gap, training must be coupled with internships that offer relevant, on-ground experience to enhance employability,” said Sameer Garde - President India and SAARC – Cisco.

“Additionally, businesses must invest in specialized cybersecurity teams to develop cyber-resilience capabilities critical for the digital future,” added Garde.

Introduction to Cyber Security



These help me in developing the skills necessary to do the following:

- Understand the importance of safe on-line behaviours.
- Describe the different types of malware and attacks
- Describe the protection strategies used organizations against attacks
- The main element of Cyber Security is the use of authentication mechanisms. For example, a username identifies an account that a user wants to access, while a password is a mechanism that proves the user is who he claims to be.

The Need for Cybersecurity

A steady rise in cybercrime highlights the flaws in devices and services we've come to depend on. This concern forces us to ask what cyber security is, why it's essential, and what to learn about it.

In this I get to know what cybersecurity is and why the demand for cybersecurity professionals is growing. It explains what our online identity and data is, where it is, and why it is of interest to cyber criminals.

Confidentiality, integrity and availability, known as the CIA triad, is a guideline for information security for an organization. Confidentiality ensures the privacy of data by restricting access through authentication encryption. Integrity

assures that the information is accurate and trustworthy. Availability ensures that the information is accessible to authorized people.

It explained why the demand for cybersecurity professionals will only continue to increase. The content explains why our personal online identity and data is vulnerable to cyber criminals. It gives some tips on how you can protect our personal online identity and data. I also get to know: - what it is, where it is, and why it must be protected. It explained who the cyber attackers are and what they want. Cybersecurity professionals must have the same skills as the cyber attackers. Cybersecurity professionals must work within the bounds of the local, national and international law. Cybersecurity professionals must also use their skills ethically.



Attacks, Concepts and Techniques

A cyber-Attack is an unauthorized attempt to access a computer system to either size, modify, or steal data.

I explore the different methods that cybercriminals use to launch an attack.

Understanding what these are and how they work is the best way to protect ourselves. So, let's make sure you know what you're up against.

I get to know the ways that cybersecurity professionals analyse what has happened after a cyberattack. I able to get to know about security software and hardware vulnerabilities and the different categories of security vulnerabilities.

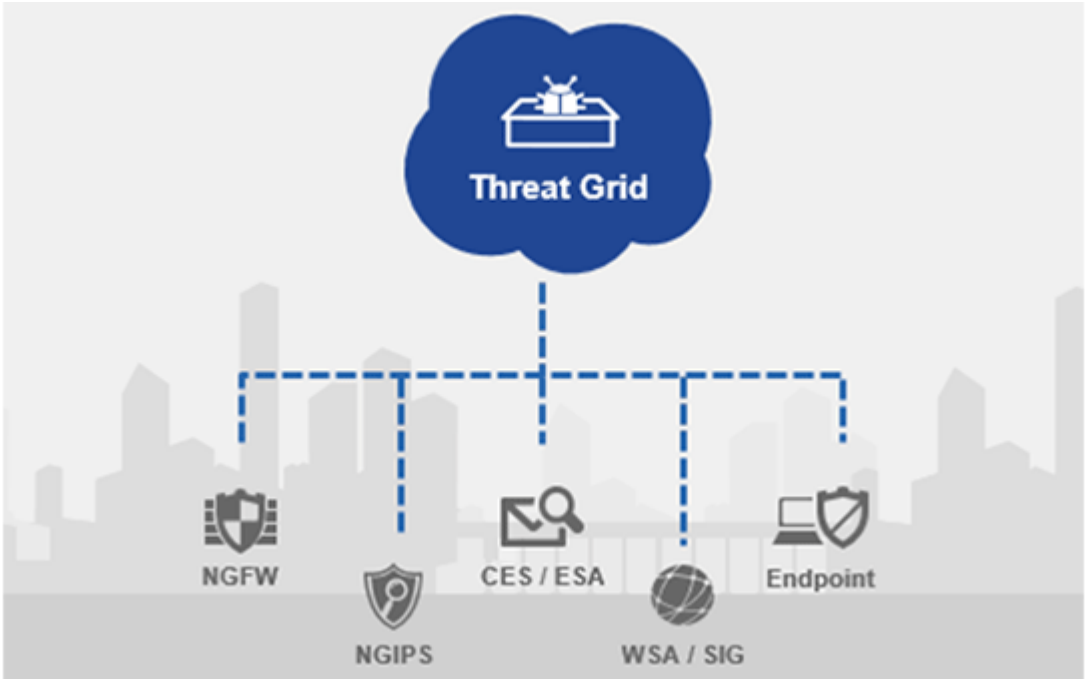
The different types of malicious software (known as malware) and the symptoms of malware are discussed. The different ways that attackers can infiltrate a system is covered, as well as denial of service attacks. Most modern cyberattacks are blended attacks. Blended attacks use multiple techniques to infiltrate and attack a system. When an attack cannot be prevented, it is the job of cybersecurity professional to reduce the impact of that attack.

Protecting Your Data and Privacy

In this I focuses on our personal devices and our personal data. It includes tips for protecting your devices, creating strong passwords and safely using wireless networks. It also discusses maintaining our data securely. Our online data is worth something to cyber criminals. Authentication techniques to help you maintain your data securely. Enhances the security of your online data with tips about what to do and what not to do online.

Protecting the Organization

In this I get to know about some of the technology and processes used by cybersecurity professionals when protecting an organization's network, equipment and data. First, it briefly covers the many types of firewalls, security appliances, and software that are currently used, including best practices. Next, botnets, the kill chain, behaviour-based security, and using NetFlow to monitor a network. The third section discusses Cisco's approach to cybersecurity, including the CSIRT team and the security playbook. the tools that cybersecurity professionals use to detect and prevent network attacks.



Will Our Future Be in Cybersecurity

In this I get to know the legal and ethical issues that arise when working in cybersecurity. Also, educational and career paths for cybersecurity. There are educational paths towards certifications that you may wish to pursue with the Cisco Networking Academy. Some of these certifications are prerequisites to Specialization Certificates in many areas of networking, including cybersecurity. The Networking Academy Talent Bridge page (netacad.com under Resources) provides good information to help you write a great résumé and prepare for a job interview. It also contains listings for Cisco and Cisco Partner jobs.

Conclusion: -

- Learn what cybersecurity is and its potential impact to you.
- Understand the most common threats, attacks and vulnerabilities.
- Gain insights for how businesses protect their operations from attacks.
- Find the latest job trends and why cybersecurity continues to grow.

Cybersecurity Essentials

Cybercriminals are constantly finding new ways to exploit vulnerabilities in systems and networks, often with the intention of stealing sensitive information and/or money.

As a cybersecurity professional, you must be able to consider the possibility of a cyber-attack occurring (a threat), recognize what makes a target susceptible to an attack (a vulnerability) and understand the different types of cyber-attacks and their effects.

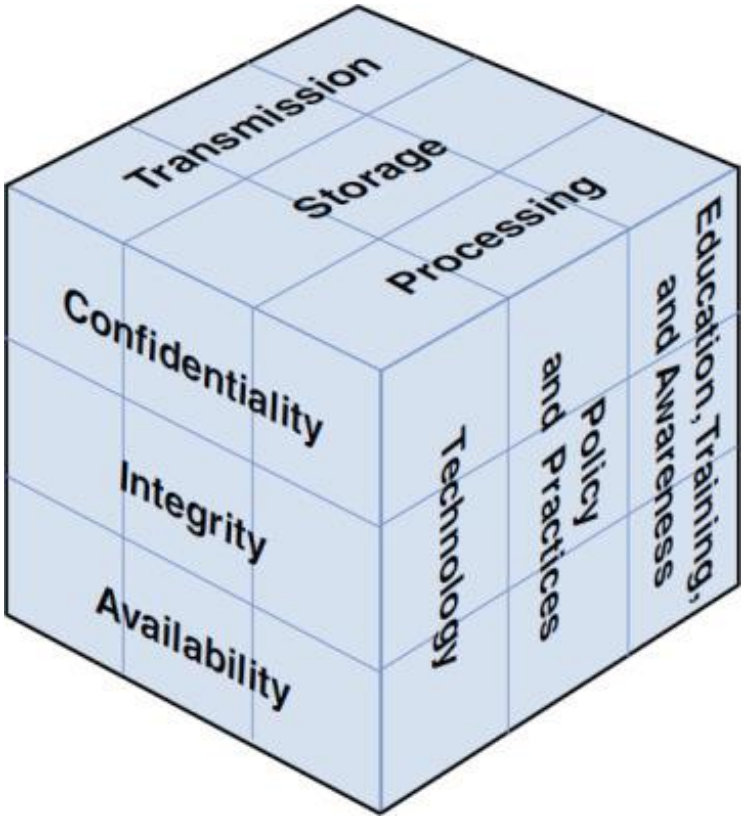
Cybersecurity Threats, Vulnerabilities and Attacks

Cyber threats may be launched to create disruption, cause damage, or to steal data, money, intellectual property, or other sensitive information. Vulnerabilities are gaps or weaknesses in an IT environment that can be exploited by attackers.

A cybersecurity threat is a deliberate and malicious attempt by an individual or company to gain unauthorized access to another person's or company's network to damage, disrupt, or steal information technology assets, intellectual property, computer networks, or any other form of sensitive data.



The Cybersecurity Cube



Cybersecurity cube is a tool developed to help manage the protection of networks, domains, and the Internet. The *first dimension* of the Cybersecurity Cube is the *three principles of information security*. The *second dimension* identifies the *three states of information or data*. The *third dimension* of the cube identifies the *expertise required to provide protection*. All three dimensions representing areas that must be addressed to secure the information system.

First Dimension: - The first dimension is what we called the *Principles of Cybersecurity*. This principle is also commonly referred to as the **CIA Triad**.

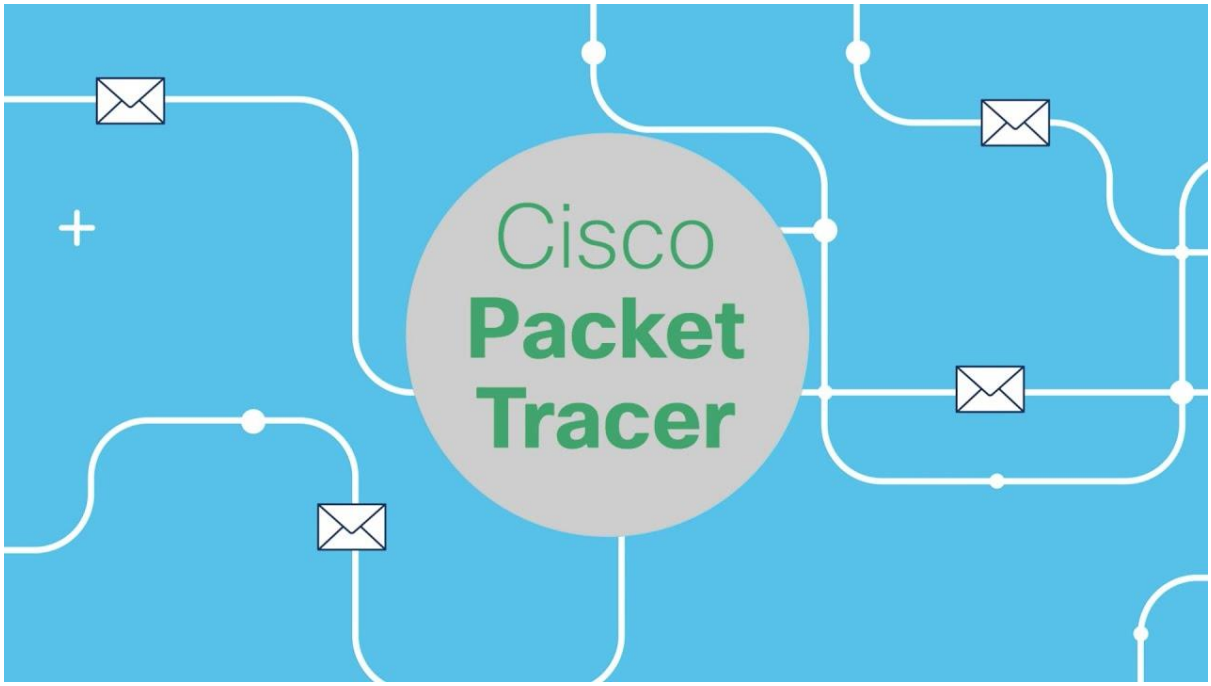
Second Dimension: - The next dimension is the Information States or the States of Data, cybersecurity cubes focus on the problem of protecting data in three possible states. The possible state of data is: Storage, transmission, process.

Third Dimension: - This dimension is often called cybersecurity safeguards. Three layers of cybersecurity safeguards are: technology, education, training.

Conclusion: -

- Understand security controls for networks, servers and applications.
- Learn valuable security principals and how to develop compliant policies.
- Implement proper procedures for data confidentiality and availability.
- Develop critical thinking and problem-solving skills using real equipment and Cisco Packet Tracer.

Introduction to Packet Tracer



With Cisco Packet Tracer, I watched how data travels across your network using Simulation Mode, or practice setting up your own device rack and cables in Physical Mode. Once I was ready, expanded my simulations to analyze network traffic, add Internet of Things (IoT) devices, integrate Python code, or practice network automation skills. Cisco Packet Tracer enables a suite of tutored activities that give us hints along the way, if we want them. Packet Tracer Tutored Activities are currently available for select courses.

Cisco Packet Tracer is Cisco's simulation software. It can be used to create complicated network typologies, as well as to test and simulate abstract networking concepts. It acts as a playground for you to explore networking and the experience is very close to what you see in computer networks.

Who uses Packet Tracer?

- Practice building simple and complex networks
- Visualize how a network works
- Practice rack, stack, and cabling skills in the virtual lab

- Integrate IoT devices, Python code, or network automation

Conclusion: -

- Practice building simple and complex networks
- Visualize how a network works
- Practice rack, stack, and cabling skills in the virtual lab
- Integrate IoT devices, Python code, or network automation
- Most importantly, Packet Tracer **helps students and instructors create their own virtual "network worlds" for exploration, experimentation, and explanation of networking concepts and technologies**.

Overview of internship experience

During my internship experience with CISCO, I was able to develop my skills. There were many Industries session organized by CISCO.

CISCO have a great line-up of speakers and a solid agenda, which will be a very exciting learning opportunity for us. I learned and get inspired and engage with industry leaders during the event.

On post complete these are my badges: -

1) https://www.credly.com/badges/3d3aa480-065b-49bb-8d3c-c1165a8470c6/public_url



2) https://www.credly.com/badges/9088c62c-b8a2-415e-9b1e-794079fd40b7/public_url

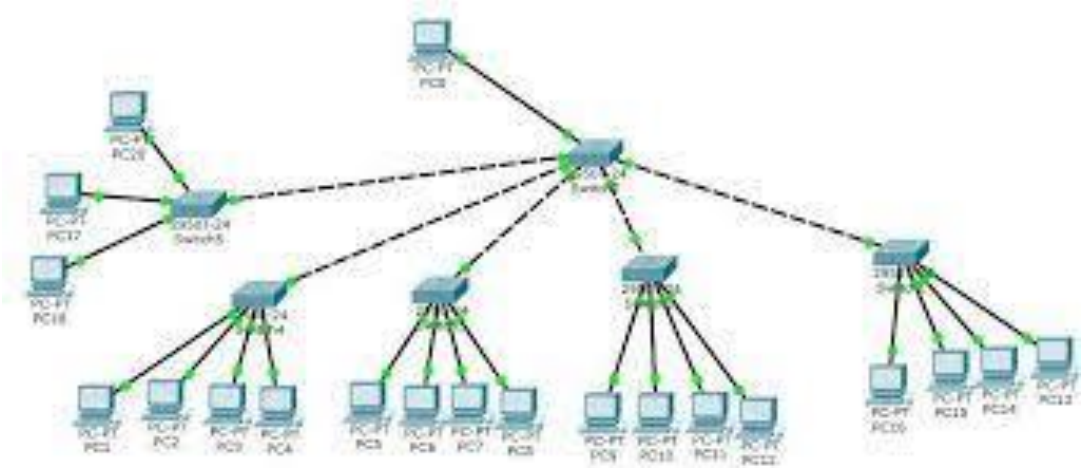


3) https://www.credly.com/badges/299067ee-bdfc-45e2-91f7-dd2dd916764e/public_url



MY PROJECT

As next step I have to do a hands-on activity to “Design Secure Network for your institution” wherein I have to study network of your institution and create a logical topology on Packet Tracer.



- In this project I learned about the different connection between different network devices.
- Learned the usage of different security tools in network.
- Hands-on knowledge on the simulation tool - Packet Tracer.
- Configuration of various types of routers, switches and other network-related connections.
- Also, designed a network for college institution.